Lab No. 08:

Shaping Tools (Weld, Trim, Intersect, and Combine)

Objective:

Understand the fundamental operations of vector shape manipulation tools in CorelDRAW (*Weld*, *Trim*, *Intersect*, *Combine*).

THEORETICAL BACKGROUND / CONCEPTS

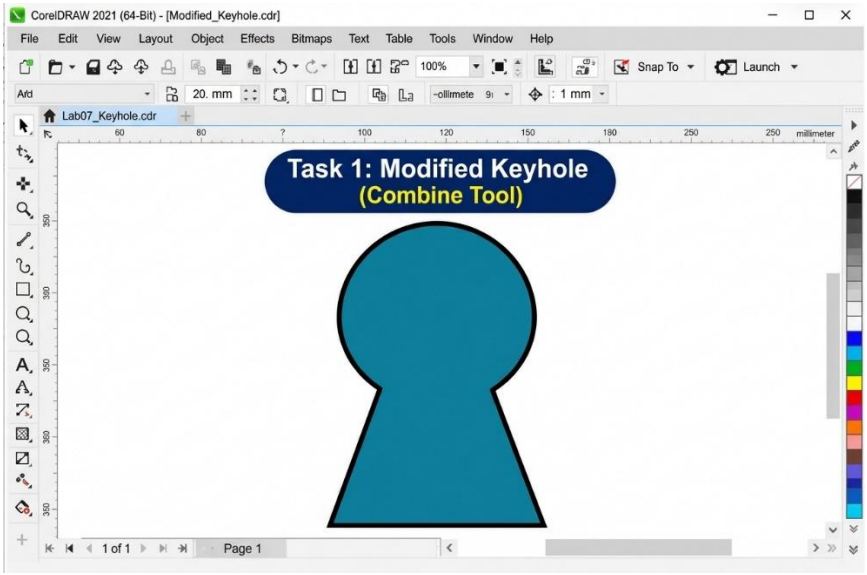
In vector graphic design software like CorelDRAW, complex shapes are created by combining simple primitive shapes using Boolean-like operations known as **Shaping Tools**.

- **Weld:** Joins two or more overlapping objects into a single object with a unified continuous outline. All overlapping inner paths and boundaries are eliminated.
- **Trim:** Uses one selected object as a cutter to remove the overlapping area from another target object.
- **Intersect:** Creates a brand-new independent object from the overlapping common area of two or more selected shapes, leaving the original shapes intact.
- **Combine (Ctrl + L):** Combines selected objects into a single curve object. Where paths overlap, transparent holes/openings are created automatically.

PRACTICAL TASKS

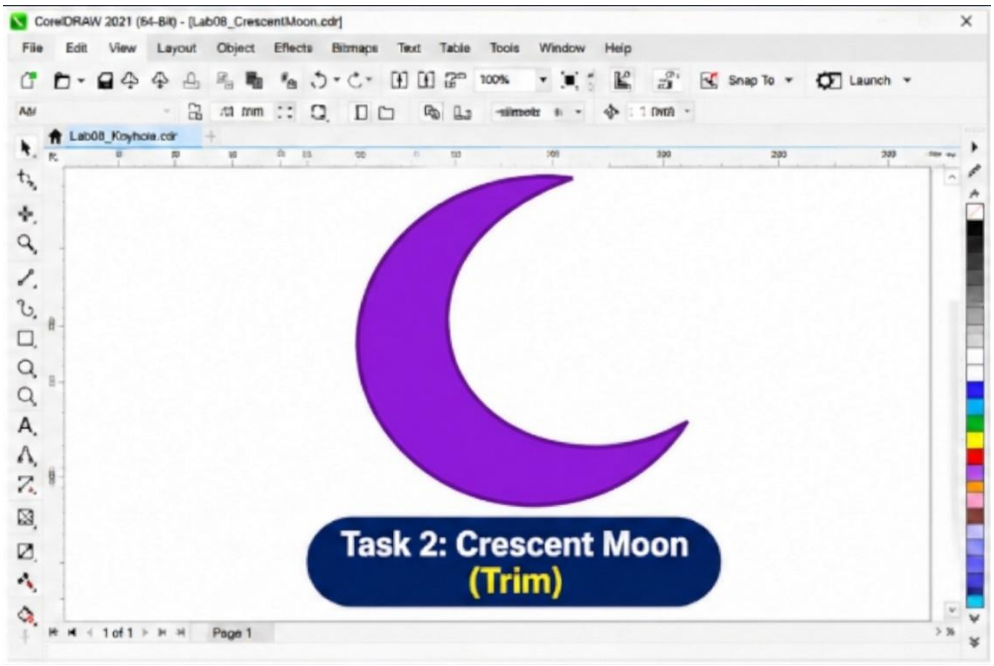
Task 1: Create a Basic Keyhole Shape (Using Combine)

- **Task Description:** Draw a circle and a vertical rectangle overlapping at the bottom, then use the **Combine** tool to form a keyhole.
- **Steps to Execute:**
    1. Select the **Ellipse Tool (F7)** and draw a circle.
    2. Select the **Rectangle Tool (F6)** and draw a narrow vertical rectangle overlapping the bottom portion of the circle.
    3. Select both shapes simultaneously.
    4. Go to **Object > Combine** (or press **Ctrl + L**).
    5. Observe the transparent keyhole opening and apply a fill color.
- **Tool Used:** Combine
- **Tool Purpose:** Merges objects into a single curve layer where overlapping regions become transparent holes.



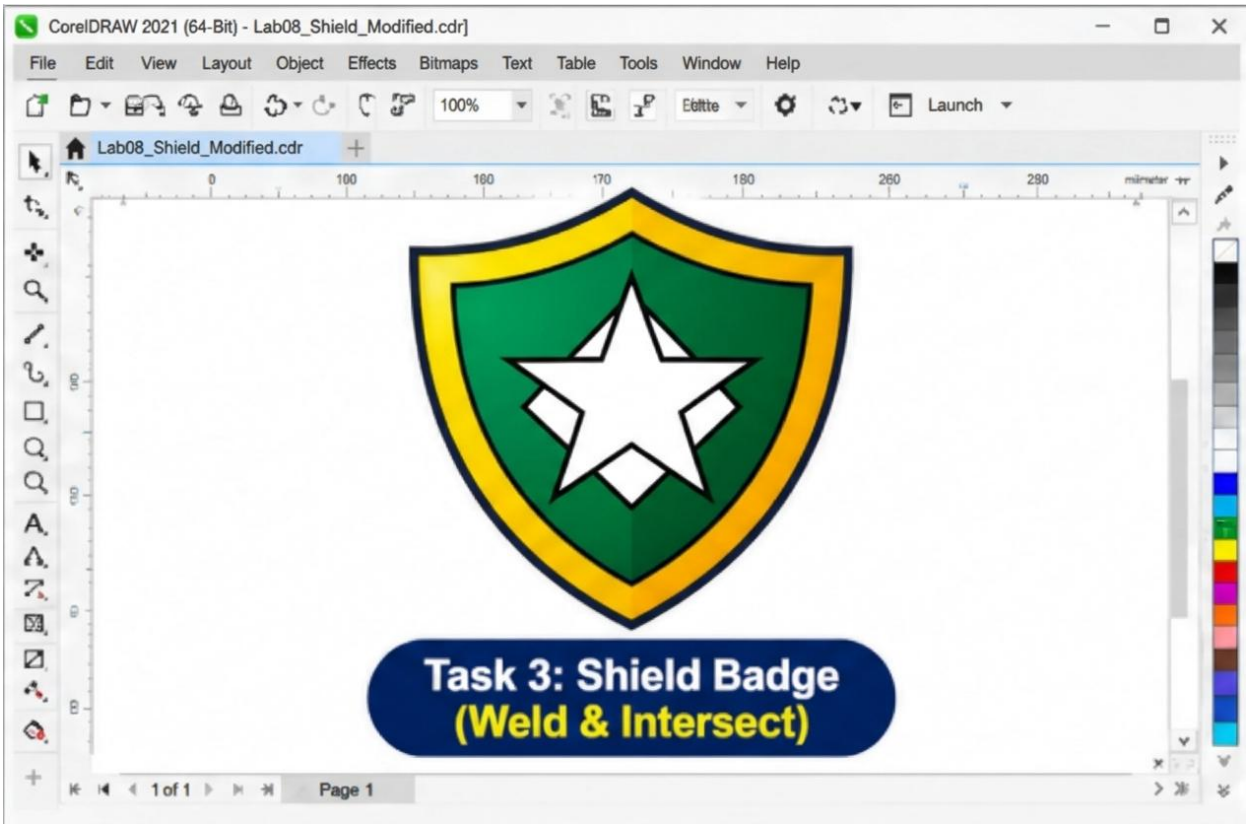
Task 2: Create a Crescent Moon (Using Trim)

- **Task Description:** Create a crescent moon shape using two overlapping circles.
- **Steps to Execute:**
    1. Draw a large circle using the **Ellipse Tool (F7)**.
    2. Duplicate the circle (**Ctrl + C**, then **Ctrl + V**) and offset it slightly over the original circle.
    3. Select both circles using the **Pick Tool**.
    4. On the Property Bar, click **Trim**.
    5. Move the cutting circle aside to reveal the trimmed crescent shape.
- **Tool Used:** Trim
- **Tool Purpose:** Cuts away overlapping portions of an underlying shape using an upper target shape.



Task 3: Create a Shield or Badge Shape (Using Weld & Intersect)

- **Task Description:** Build a heraldic shield structure using a rectangle and a triangle, then extract an inner emblem using Intersect.
- **Steps to Execute:**
    1. Draw a rectangle and an inverted triangle; align the triangle at the bottom of the rectangle.
    2. Select both shapes and click **Weld** to create a unified outer shield outline.
    3. Place a smaller shape (e.g., star or circle) over the center of the shield.
    4. Select both the shield and inner shape, then click **Intersect** to extract the overlapping badge emblem.
- **Tools Used:** Weld & Intersect
- **Tool Purpose: Weld** merges outer boundaries into a single shape, while **Intersect** creates a new object out of the overlapping central region.



Task 4: Build a Teddy Bear Character (Comprehensive Exercise)

- **Task Description:** Construct a complete teddy bear character by applying all four shaping operations (*Weld*, *Trim*, *Combine*, *Intersect*) in sequence.
- **Steps to Execute:**
    1. **Stage 1 (Combine & Assembly):** Draw circles for the head, ears, muzzle, torso, and limbs. Use **Combine (Ctrl + L)** for inner ear details.
    2. **Stage 2 (Weld):** Select the head, ear bases, torso, and arm/leg shapes, then click **Weld** to form the bear's primary body silhouette.
    3. **Stage 3 (Trim):** Position circular shapes over the ear areas and muzzle section, then click **Trim** to shape the inner curves cleanly.

4. **Stage 4 (Intersect & Refinement):** Draw ellipses over the feet/paws and apply **Intersect** to generate precise, aligned footpads. Apply color fills to complete the bear design.

- **Tools Used:** Weld, Trim, Intersect, Combine
- **Tool Purpose:** Combines multiple operations to build complex, multi-layered vector character illustrations from basic geometric primitives.

